

Gut Inflammation:

Manifestations – Mediators – Mitigators



SWATHI RAO, PA-C

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Swathi Rao, PA-C, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

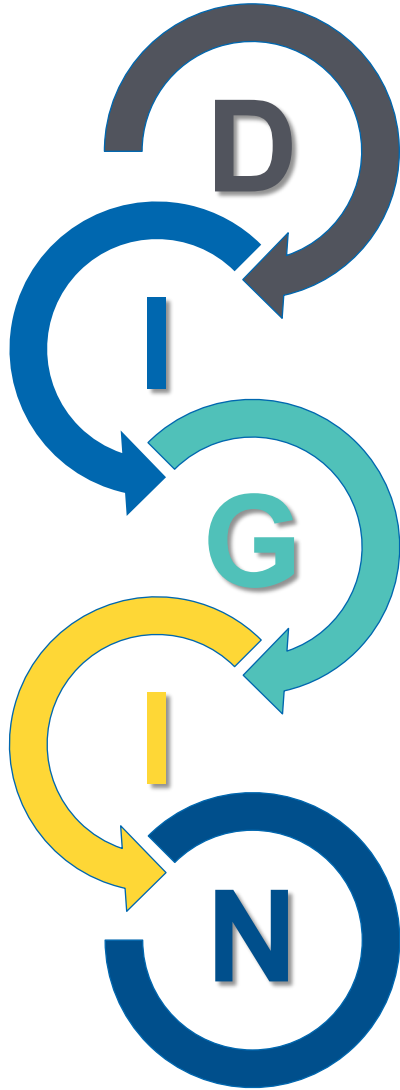
1. Contrast the normal physiology of gut immune function and regulation and the pathological mechanisms of gut immune dysregulation.
2. List common clinical findings, antecedents, triggers, and mediators of gut immune dysregulation.
3. Explain how gut immune dysregulation may serve as a mediator of extraintestinal disease and/or dysfunction

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Document clinical findings of gut inflammation on the Matrix and Timeline. Identify and elicit the antecedents, triggers, and mediators of these issues.
2. Identify extraintestinal diseases and dysfunctions that may be mediated by gut immune dysregulation.

The DIGIN Mnemonic of Gut Dysfunctions



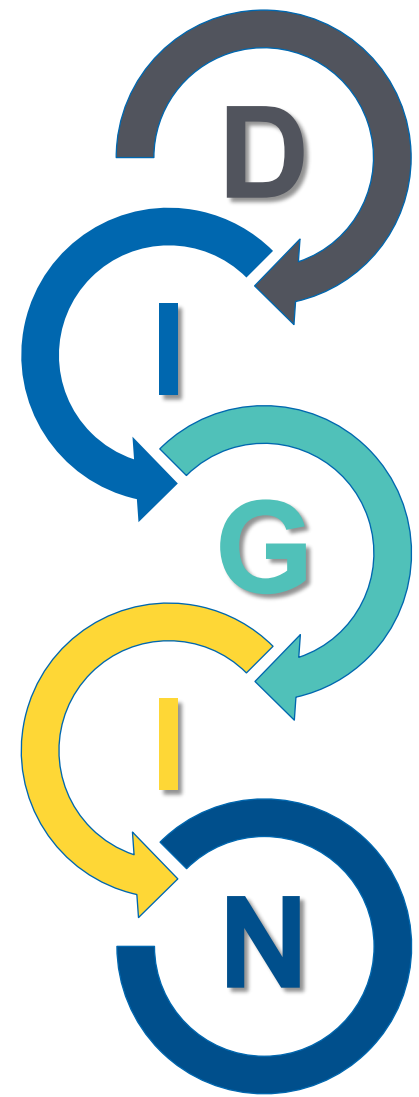
Digestion/Absorption

Intestinal Permeability

Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Clinical Focus

- Definitions and terminology



Terminology

- **Gut inflammation** refers to the immune system's chronic, dysregulated response within the gastrointestinal tract, often accompanied by maldigestion, increased intestinal permeability, altered gut microbiota, and/or GI immune system activation.
- Gut inflammation can be either a mediator or an effect of systemic immune dysregulation or systemic inflammation.

Terminology

- **Local gut inflammation** may refer to luminal, mucosal, or brush border/epithelial inflammation.
- **GALT**: The gut-associated lymphoid tissue in the mucosa and submucosa serves as a fulcrum between local gut inflammation and systemic immune dysregulation.
- **Systemic chronic inflammation** occurs when the immune system continues to react to what it perceives as a threat. When it is a sustained response, it may contribute to a wide variety of chronic inflammatory diseases and dysfunctions, including gut inflammation.

Terminology

- **Lysozyme** is an antimicrobial peptide within the mucus layer.
- **Secretory immunoglobulin A (sIgA)** neutralizes pathogens and binds to antigens to prevent their movement across an impaired intestinal lining, preventing an immune response.

Clinical Focus

- Normal physiology of gut immune regulation and the gut-associated lymphoid tissue (GALT).



Gut Immunity Guardians

Chemical Barriers

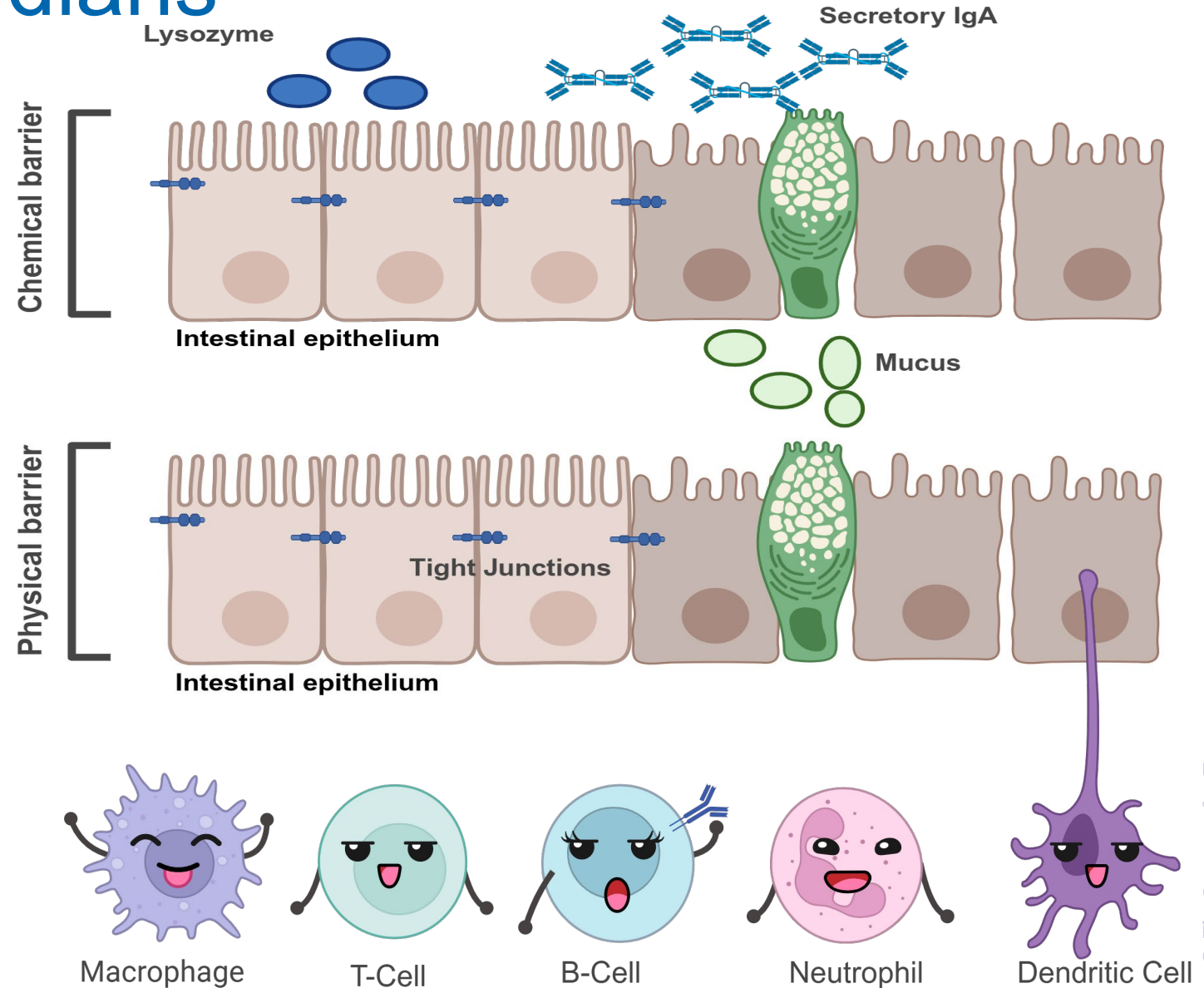
- Lysozyme, HCl
- sIgA

Physical Barriers

- Mucosa
- Tight junctions

Immunological Barriers

- Mucosa
- Sub-mucosa



Long Description: Gut Immunity Guardians

- Chemical barriers of intestinal immunity include lysozymes, H C L, and secretory I G A. Physical barriers include the intestinal mucosa and tight junctions between intestinal epithelial cells. Immune barriers in the mucosa and submucosa include macrophages, T cells, B cells, neutrophils, and dendritic cells.

Chemical Barriers

Antimicrobial peptides (AMPs): lysozyme, slgA, defensins, etc.

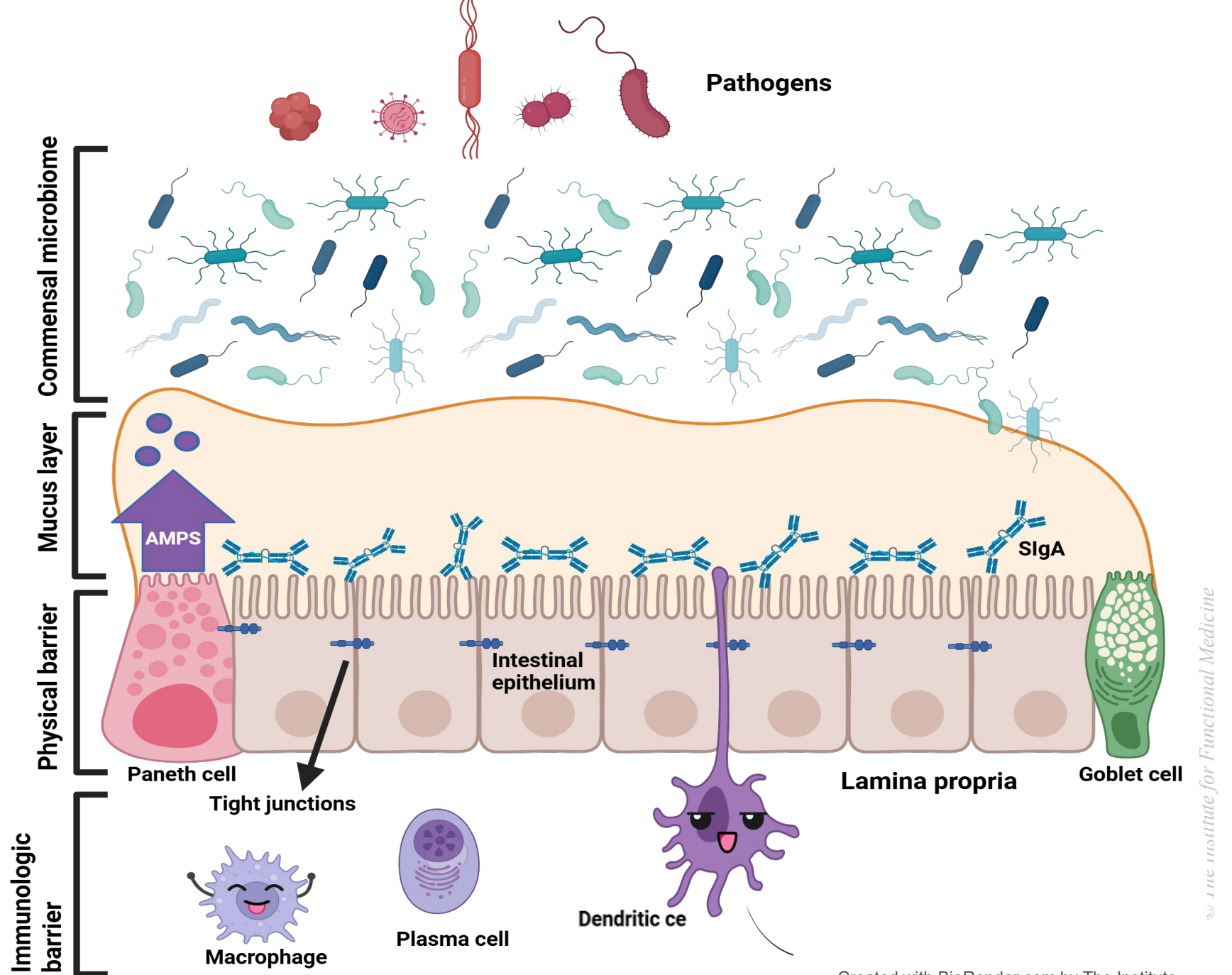
Physical Barriers

Microbiome, mucosa, tight junctions

Immunological Barriers

GALT components

See: "References: Chemical, Physical, and Immunological Barriers"



Long Description: Gut Barriers

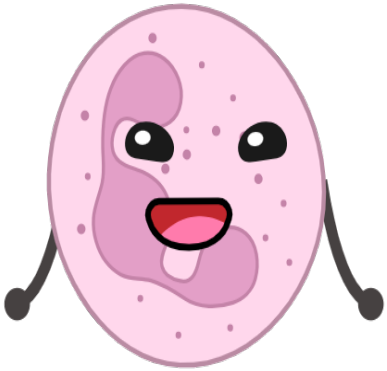
- The intestinal barrier is comprised of physical, chemical, and immunological barriers. The physical barrier includes the commensal microbiome in the intestinal lumen, a mucus layer, and an underlying monolayer of intestinal epithelium joined by tight junctions interspaced with Paneth and Goblet cells. Paneth cells secrete antimicrobial peptides such as lysozymes and defensins, which are part of the chemical barrier. Beneath the epithelium, the lamina propria contains the immunological barrier composed of macrophages, plasma cells, dendritic cells, T cells, and B cells. Secretory IgA is a component of the immunological barrier secreted into the intestinal lumen.

References

CHEMICAL, PHYSICAL, AND IMMUNOLOGICAL BARRIERS

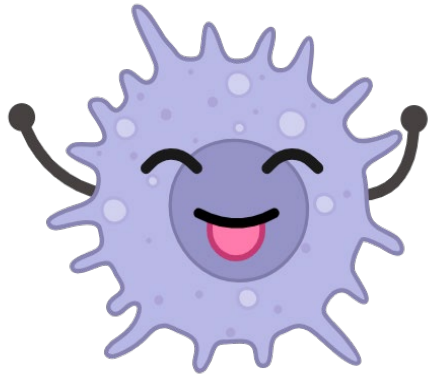
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Immunological Components of GALT



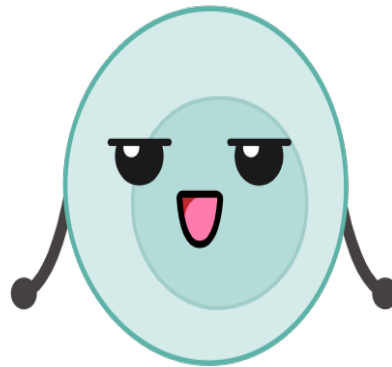
Neutrophils

Engulf and destroy bacteria and other foreign particles



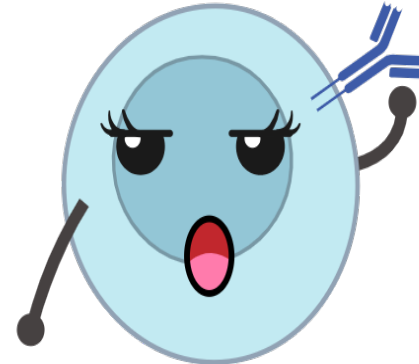
Macrophages

Pathogen and cellular debris clearance



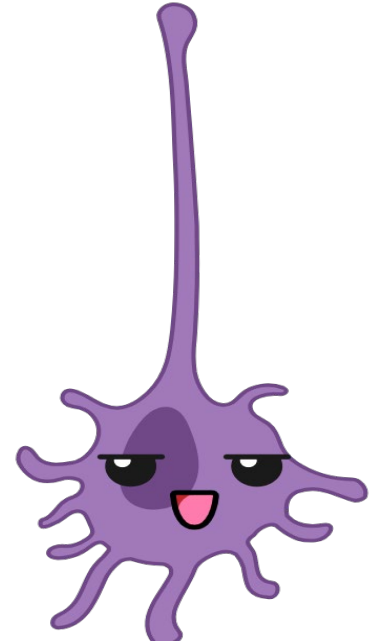
T-Cells

Protect mucosal barrier



B-Cells

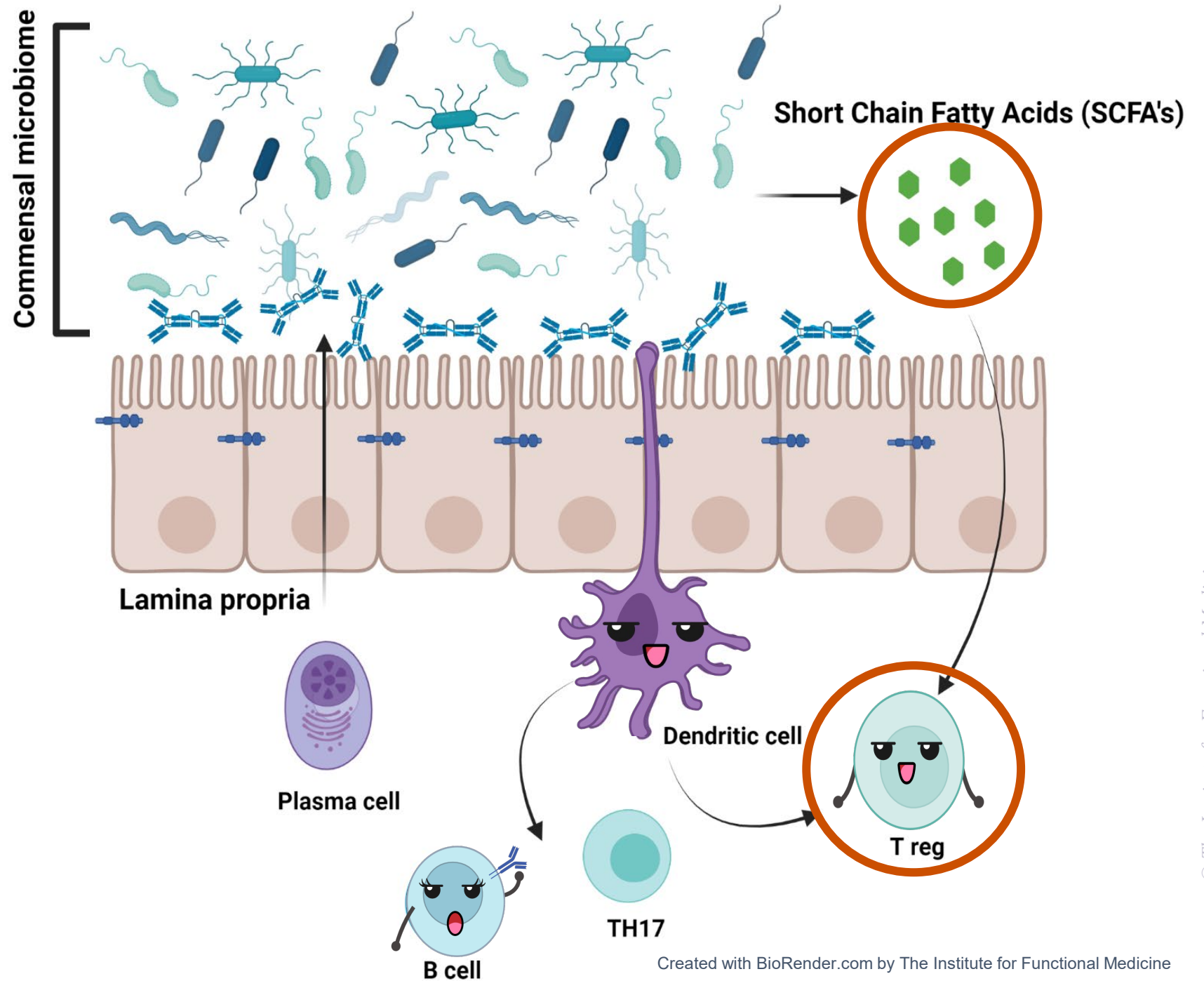
Adaptive immunity



Dendritic Cells

Present antigens to T-cells
Induce tolerance

The Microbiome Role in Gut-Immune Regulation and Defense



Long Description: The Microbiome Role in Gut-Immune Regulation and Defense

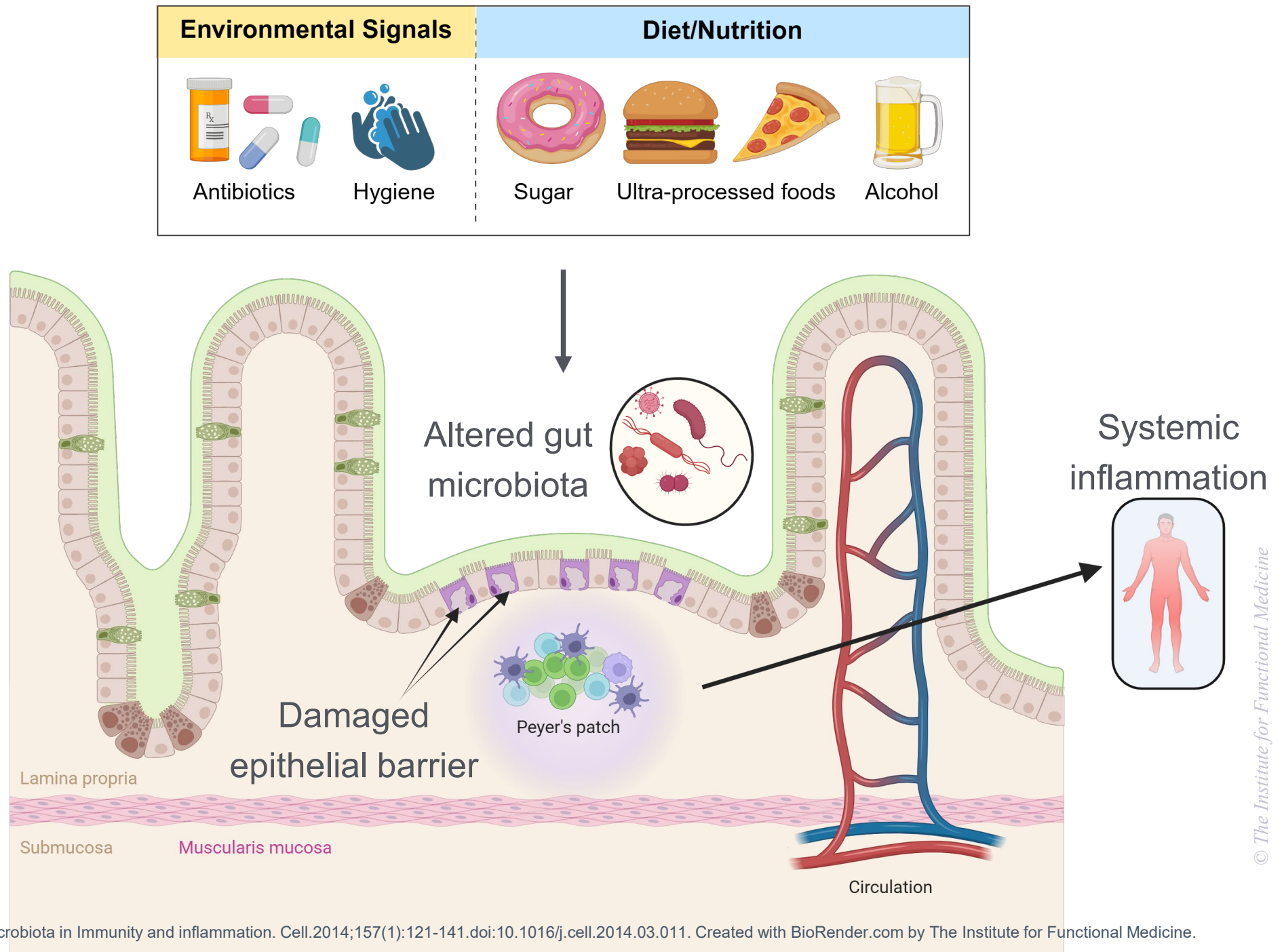
- In a healthy immune response in the gut, short chain fatty acids help to enhance T regulatory cells, supporting immune regulation.

References

THE MICROBIOME ROLE IN GUT-IMMUNE REGULATION AND DEFENSE

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GALT & Gut Inflammation: The Fulcrum Between Inflammatory Mediators and Chronic Systemic Inflammatory Diseases



Long Description: G A L T & Gut Inflammation

- Environmental signals such as antibiotic use, hygiene practices, and dietary choices such as sugar, ultra-processed foods, and alcohol are mediators of systemic inflammation through altered gut microbiota and a damaged epithelial barrier.

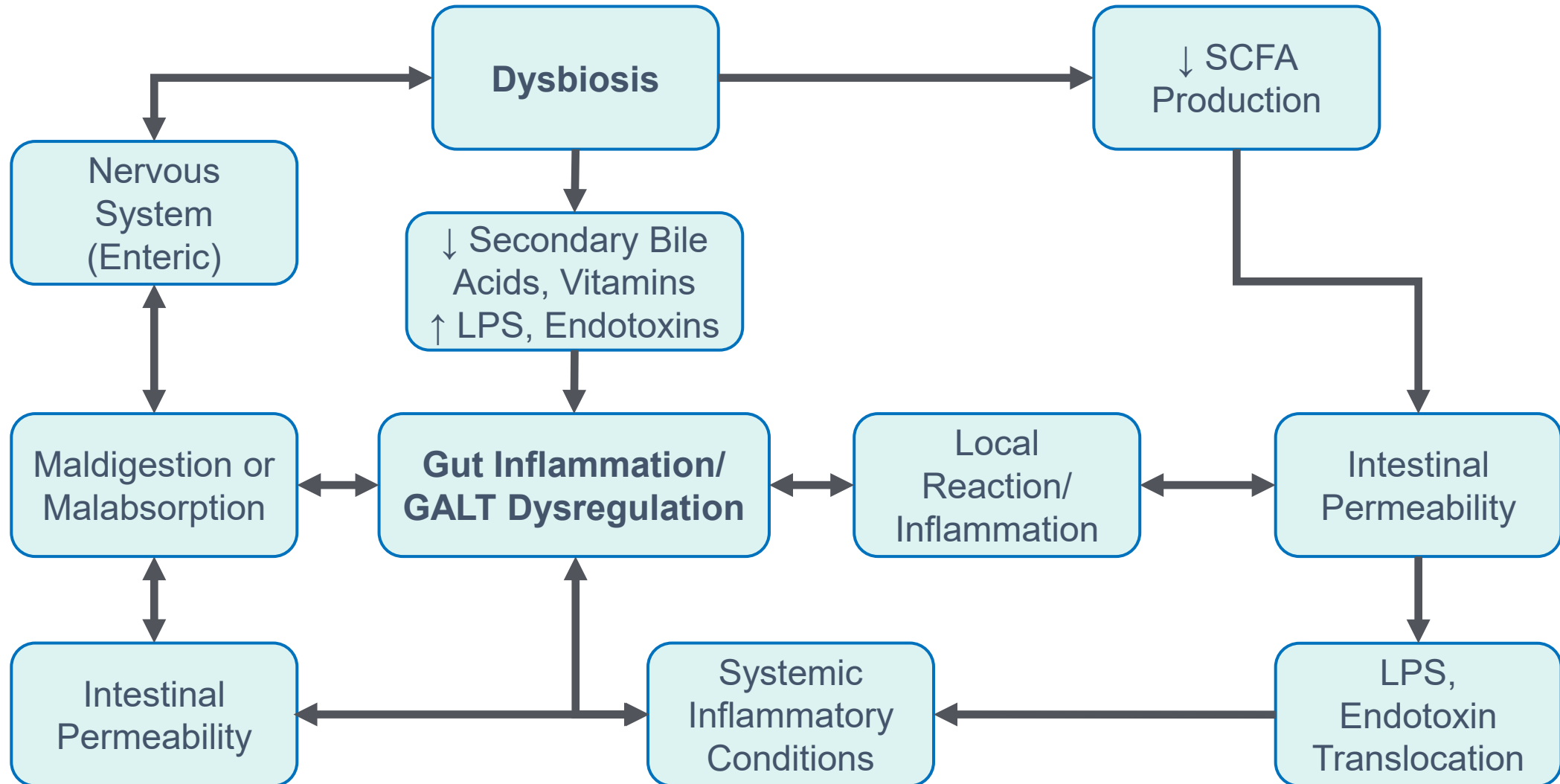
Clinical Focus

- Pathophysiological mechanisms of dysregulation of gut immune function.



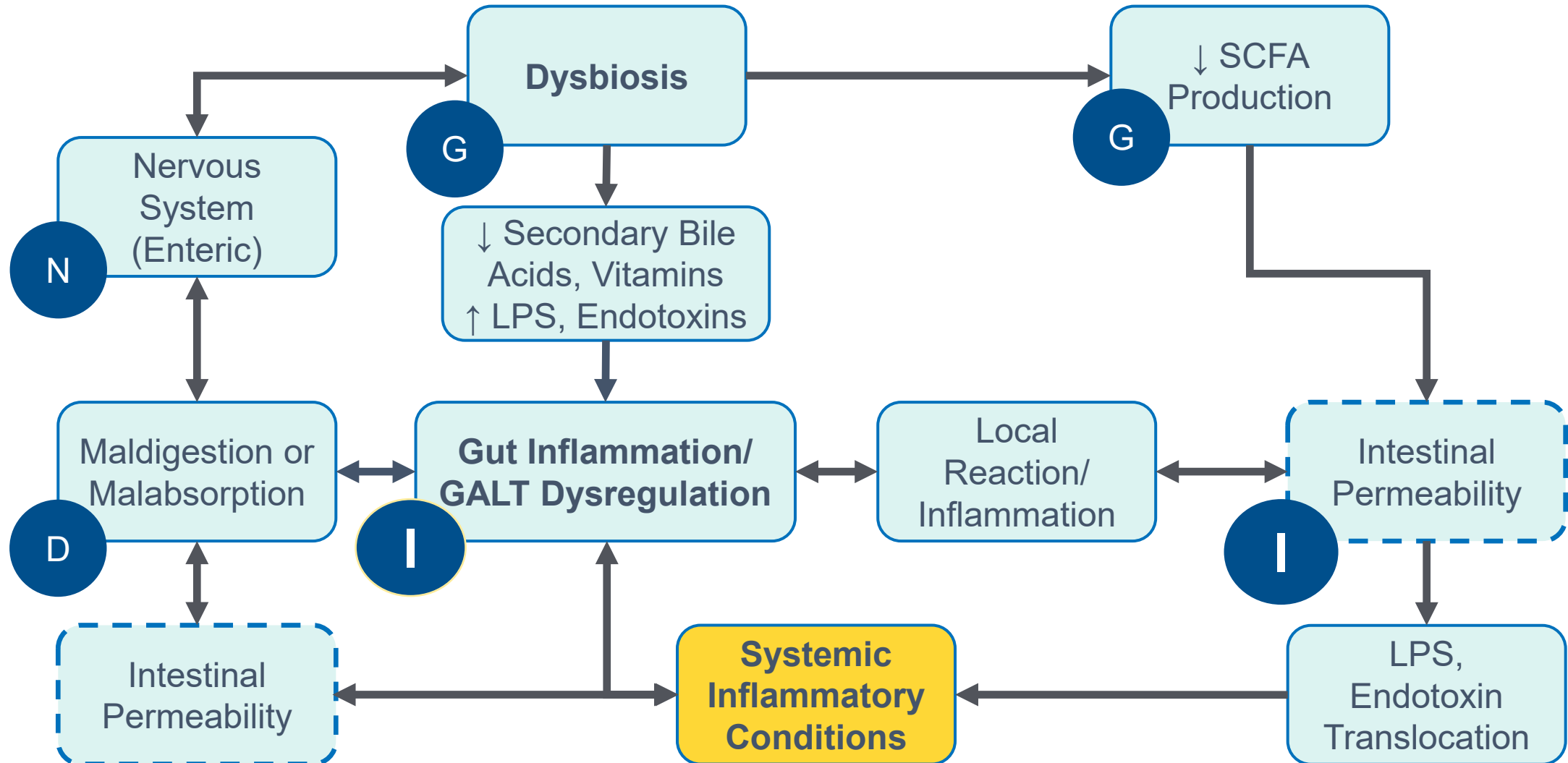
Clinical Scenarios of Gut/GALT Inflammation

Correlation With DIGIN Dysfunctions



Clinical Scenarios of Gut/GALT Inflammation

Correlation With DIGIN Dysfunctions

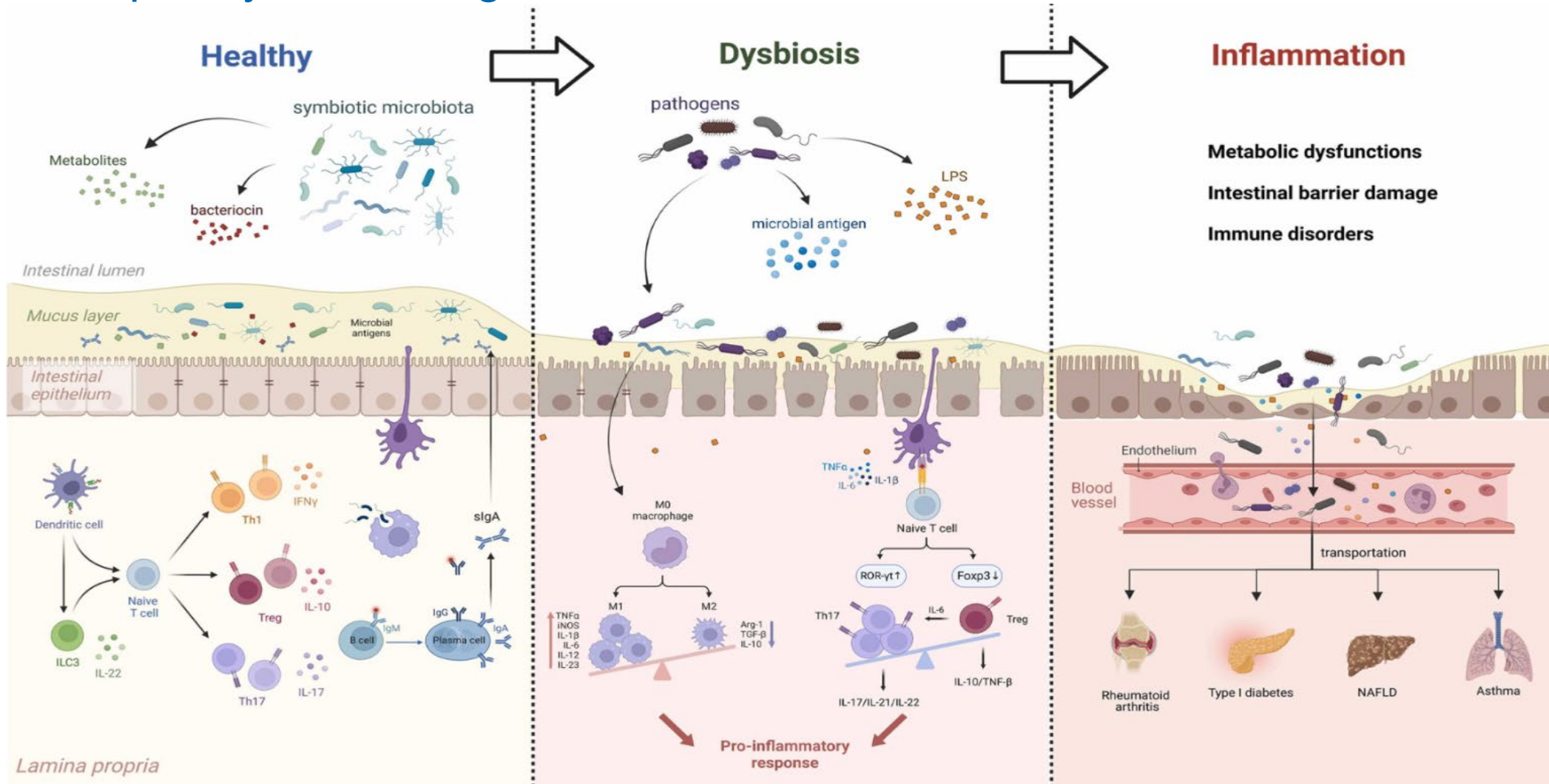


Long Description: Clinical Scenarios of Gut/G A L T Inflammation

- Recall the D I G I N dysfunctions and their mutual reinforcements that lead to systemic inflammatory conditions
 - D. Digestion, absorption
 - I. Intestinal permeability
 - G. Gut microbiota, dysbiosis
 - I. Inflammation, immune modulation
 - N. Nervous system, enteric division

Deeper Dive Into DIGIN Correlations

Next steps in your training



- Image:Zhao M, Chu J, Feng S, Guo C, Xue B, He K, Li L. Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: A review. *Biomed Pharmacother.* 2023 Aug;164:114985. doi: 10.1016/j.biopha.2023.114985. Epub 2023 Jun 11. PMID: 37311282. CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).
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Long Description: Deeper Dive Into D I G I N Correlations

- In a healthy intestinal environment, symbiotic microbiota produce bacteriocins and metabolites that promote a healthy immune response and keep pathogenic microbes in balance. In dysbiosis, pathogenic microbes secrete L P S and microbial antigens, damaging the intestinal barrier and promoting a proinflammatory response that can lead to metabolic dysfunction and immune disorders.

Clinical Focus

- Clinical findings of gut inflammation.



Representative Patient Clinical Findings

Gut Inflammation

Heartburn

Dermatitis

Gastritis

Cardiovascular Disease

GERD

Autoimmunity

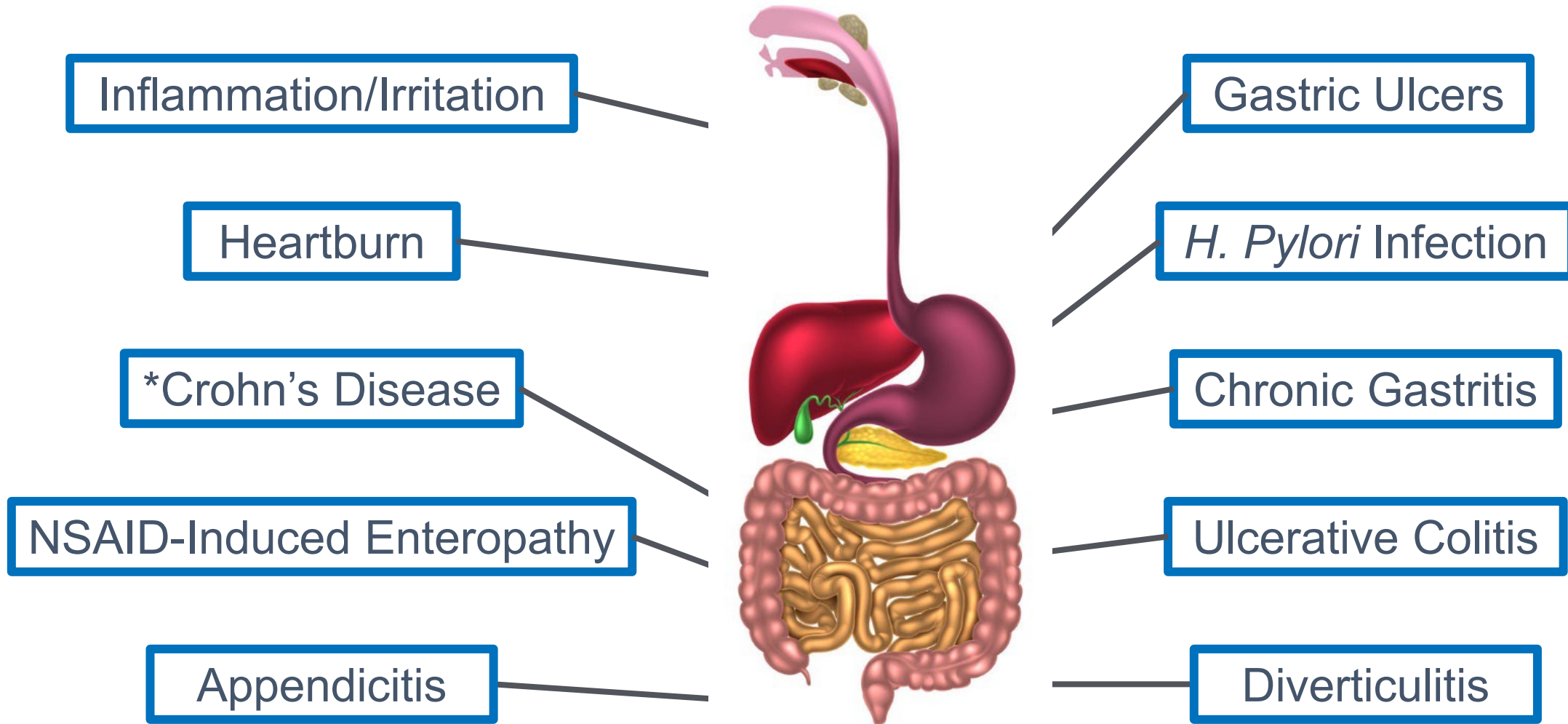
Dyspepsia

Depression/Anxiety



Gut Inflammation

Local Clinical Findings Throughout the GI Tract



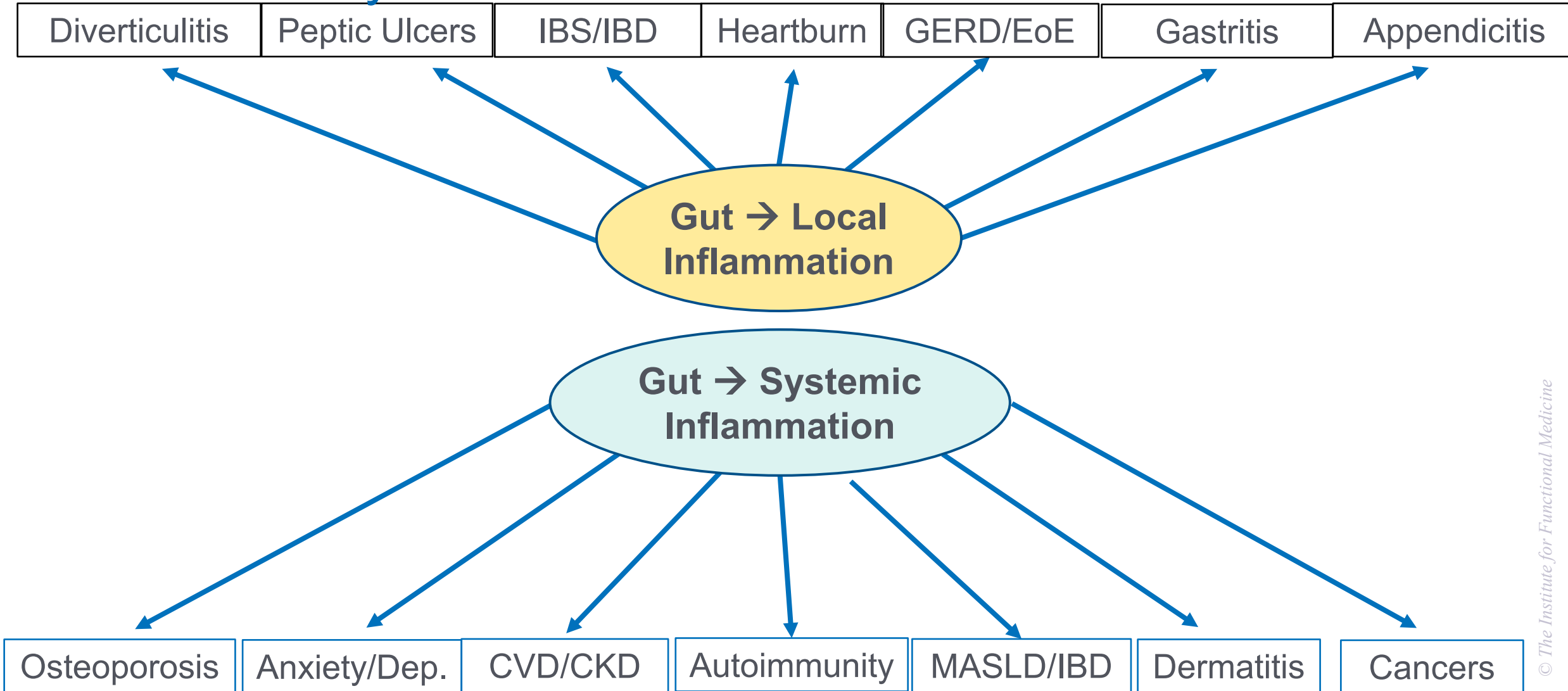
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Gut Inflammation

Local and Systemic Manifestations



See: "References: Local and Systemic Manifestations of Gut Inflammation"

Long Description: Gut Inflammation

- Local gut inflammation manifestations
 - Diverticulitis
 - Peptic ulcers
 - I B S and I B D
 - Heartburn
 - GERD and E O E
 - Gastritis
 - Appendicitis
- Systemic gut inflammation manifestations
 - Osteoporosis
 - Anxiety, depression
 - C V D and C K D
 - Autoimmunity
 - M A S L D
 - I B D
 - Dermatitis
 - Cancers

References

LOCAL AND SYSTEMIC MANIFESTATIONS OF GUT INFLAMMATION

Diverticulitis:

- Ma W, Wang Y, Nguyen LH, et al. Gut microbiome composition and metabolic activity in women with diverticulitis. *Nat Commun*. 2024;15(1):3612. doi:10.1038/s41467-024-47859-4

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Heartburn:

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LOCAL AND SYSTEMIC MANIFESTATIONS OF GUT INFLAMMATION

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Anxiety/Depression:

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CVD/CKD:

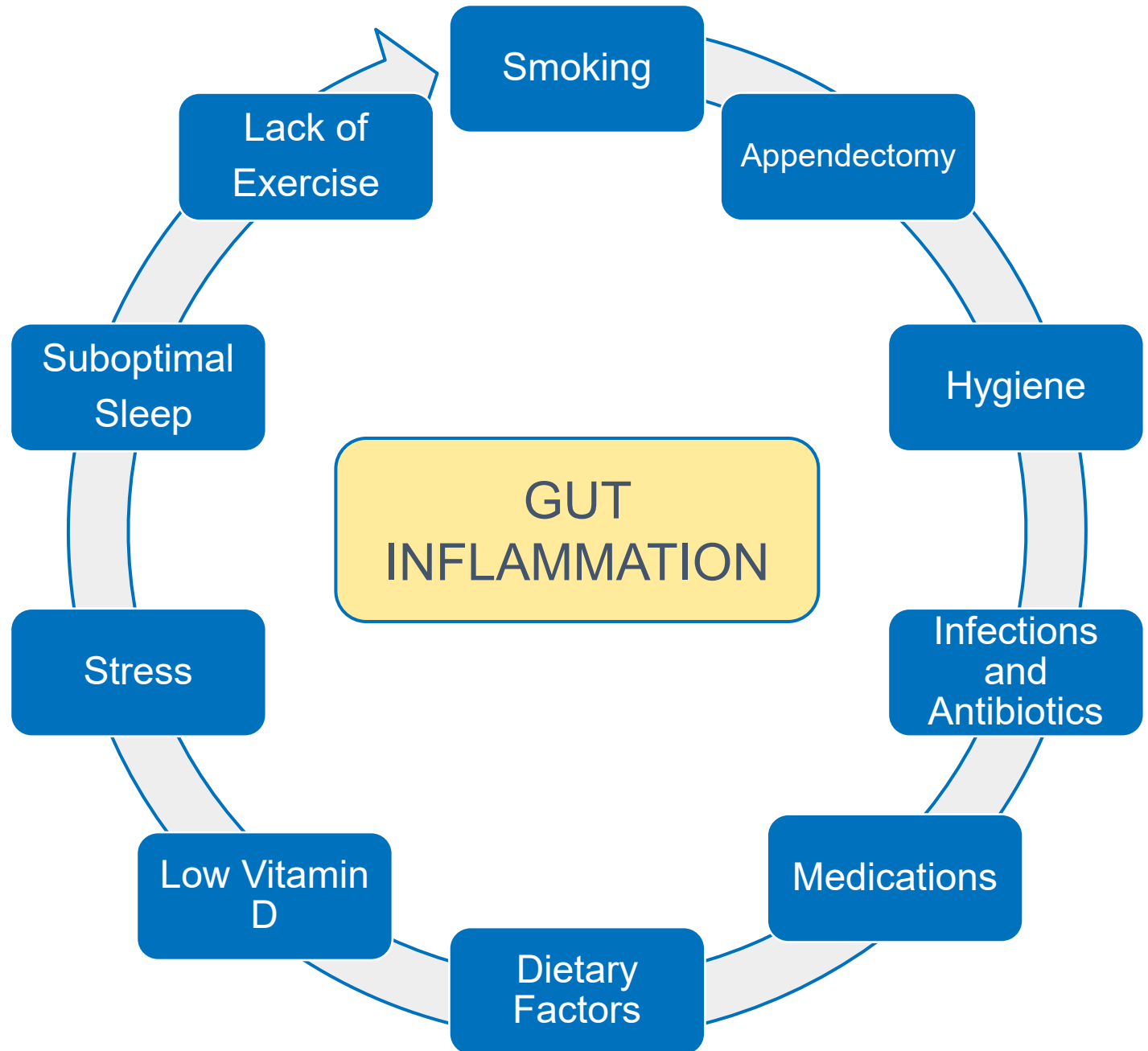
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Clinical Focus

- Antecedents, triggers, and mediators of gut inflammation.

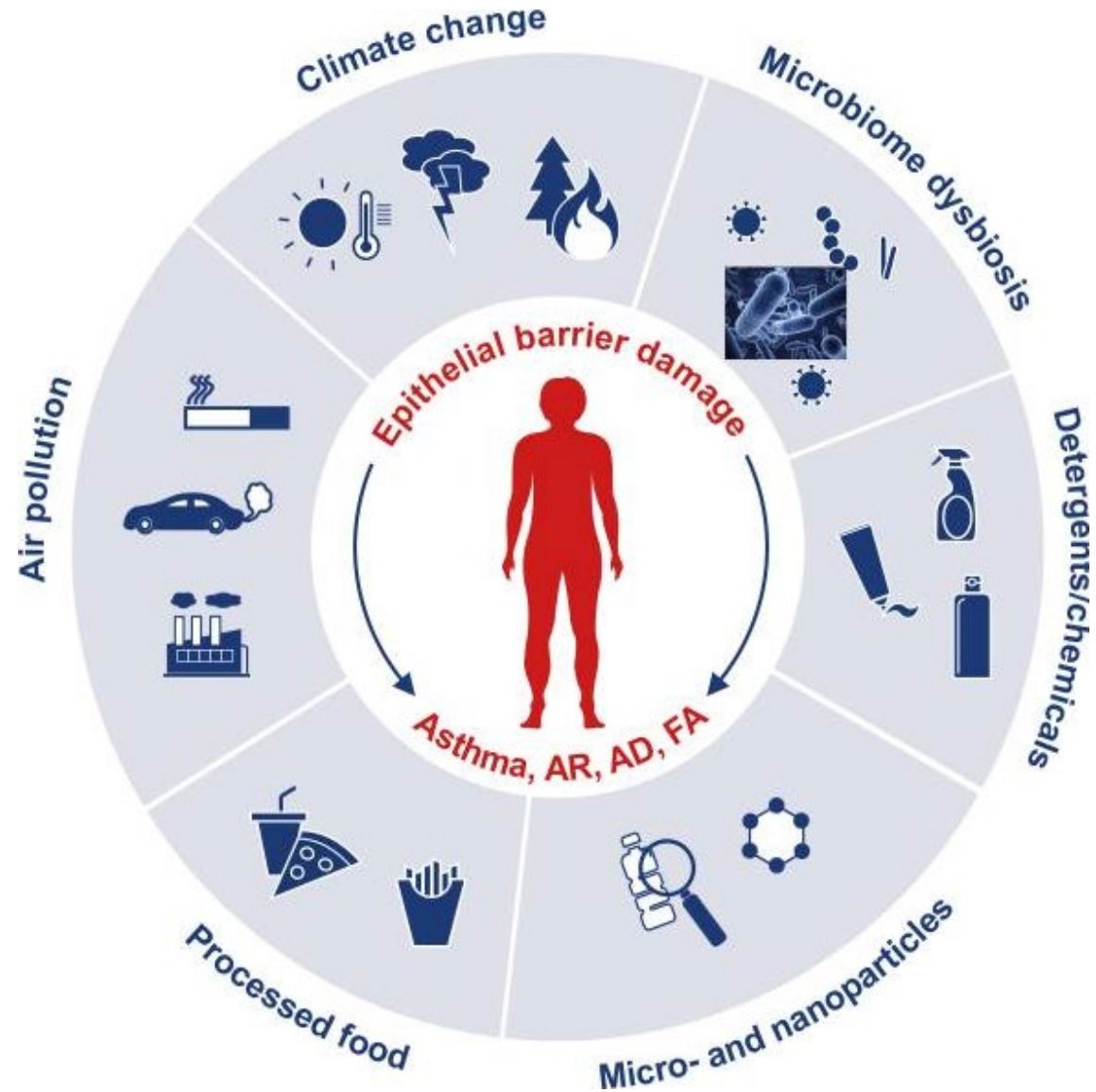


Antecedents, Triggers, and Mediators: Gut Inflammation



21st Century “Emerging” Antecedents, Triggers, and Mediators

Gut Inflammation



Long Description: 21st Century “Emerging” Antecedents, Triggers, and Mediators, Gut Inflammation

- Climate change
- Microbiome dysbiosis
- Detergents and chemicals
- Micro and nanoparticles
- Processed food
- Air pollution

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents (Inherited & Acquired)

- Bottle fed
- Genetics

Triggers & Precipitating Events

- Acute toxin exposure
- Infections

Mediators/Perpetuators

- Environmental exposures
- Medications

Modifiable Lifestyle Factors

Sleep

- Poor sleep

Physical Activity

- Sedentarism
- Overexercise

Nutrition

- Malnutrition
- Nutrient imbalance

Stressors

- Life stressors

Relationships

- Loneliness

Assimilation

- Dysbiosis
- Maldigestion
- Dysmotility

Structural Integrity

- Increased Intestinal permeability

Communication

- Gut/brain axis issues
- Adrenal dysregulation
- High sympathetic tone
- Vagal nerve

Defense & Repair

- sIgA deficiency
- Autoimmunity
- Systemic inflammation

Energy

- Oxidative stress

Mitigation & Clearance

- Toxins

Transport



Mediators of Gut Inflammation

Losol P, Sokolowska M, Hwang YK, et al. Epithelial barrier theory: the role of exposome, microbiome, and barrier function in allergic diseases. *Allergy Asthma Immunol Res.* 2023;15(6):705-724. doi:10.4168/aa.2023.15.6.705

Long Description: Mediators of Gut Inflammation

- Mediators of gut inflammation functional medicine matrix
- Antecedents
 - Bottle Fed
 - Genetics
- Triggers and precipitating events
 - Acute toxin exposure
 - Infections
- Mediators and perpetuators
 - Environmental exposures
 - Medications
- Modifiable lifestyle factors
 - Poor sleep
 - Sedentarism
 - Over-exercise
 - Malnutrition
 - Nutrient imbalance
 - Life stressors
 - Loneliness
- Clinical Imbalances and physiological dysfunctions
 - Assimilation
 - Dysbiosis
 - Maldigestion
 - Dysmotility
 - Defense and repair
 - Secretory I G A deficiency
 - Autoimmunity
 - Systemic inflammation
 - Energy
 - Oxidative stress
 - Mitigation and Clearance
 - Toxins
 - Transport
 - Communication
 - Gut-brain axis issues
 - Adrenal dysfunction
 - High sympathetic tone
 - Vagal nerve
 - Structural Integrity
 - Intestinal permeability
 - Mental, emotional, and spiritual
 - Anxious and depressed feelings

Clinical Focus

- Gut inflammation as a mediator of extra-intestinal conditions.



Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents (Inherited & Acquired)

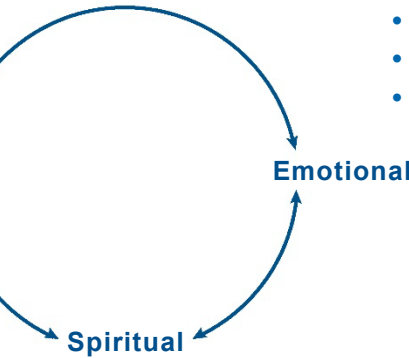
- DIGIN
- Chronic gastritis
- Crohn's disease, UC
- NSAID-induced enteropathy

Triggers & Precipitating Events

Mediators/Perpetuators

- Assimilation**
- Peptic ulcer, blunted villi
 - Osteopenia/osteoporosis
 - Sarcopenia

- Structural Integrity**
- Communication**
- Fertility challenges
 - NAFLD/MASLD
 - Metabolic disease
 - Diabetes; insulin resistance
 - Metabolic syndrome



Defense & Repair

- Autoimmunity
- Systemic inflammation
- GALT dysfunction
- Dermatitis, eczema

Energy

- Fatigue
- Neurodegenerative disease

Mitigation & Clearance

- Liver inflammation
- MASLD

Transport

- Kidney disease
- CVD

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

- Maldigestion
- Malabsorption

Stressors

Relationships

Gut Inflammation as Mediator

- Heartburn
- Irritation & inflammation by hot beverages and fungi
- Gastric ulcers
- *H. pylori* infection
- Chronic gastritis
- Crohn's disease
- NSAID-induced enteropathy
- Ulcerative colitis

Long Description: Gut Inflammation as Mediator

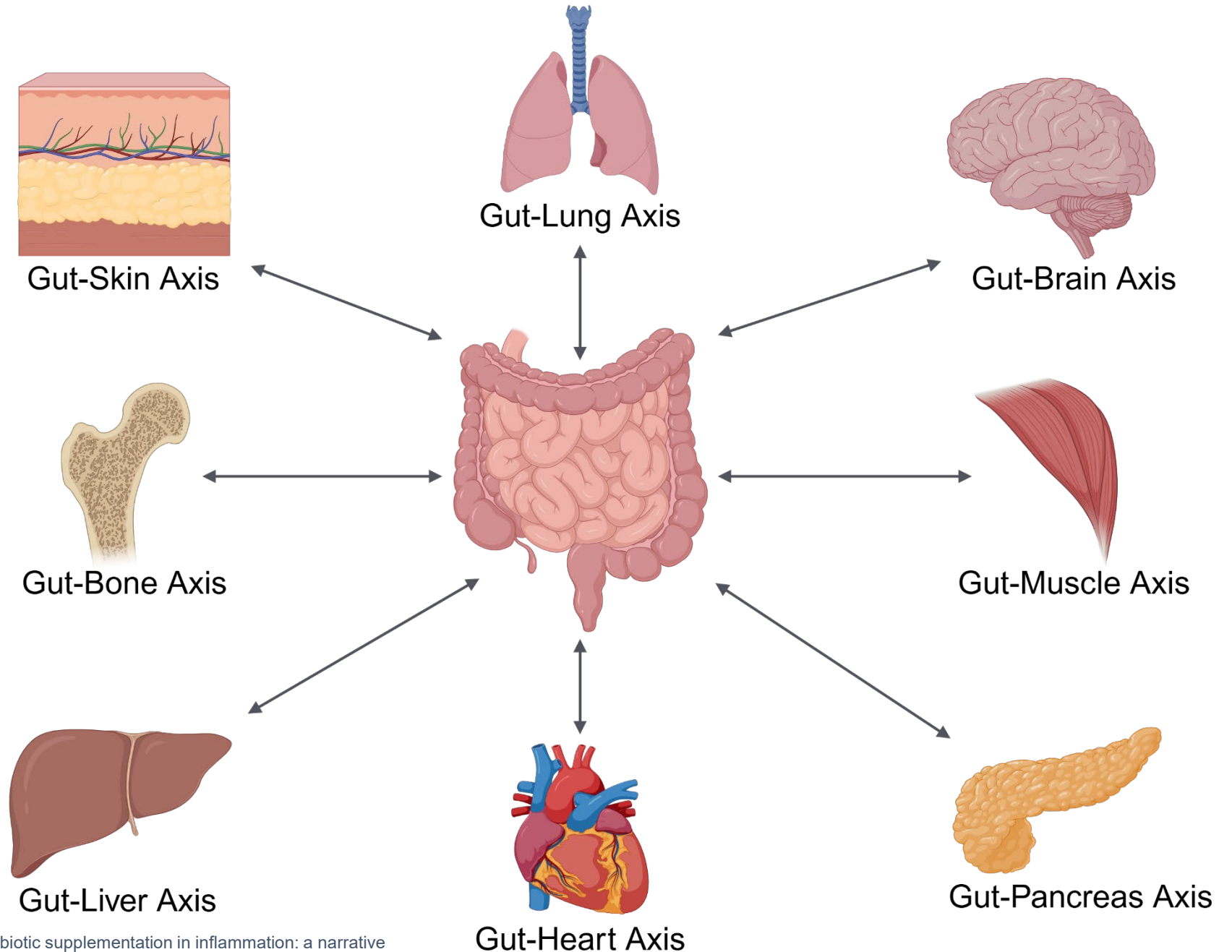
- Functional Medicine Matrix
- Gut Inflammation as a mediator
 - Heartburn
 - Irritation & Inflammation by hot beverages and fungi
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 - *H. pylori* infection
 - Chronic gastritis
 - Crohn's disease
 - NSAID-induced enteropathy
 - Ulcerative colitis
- Defense and repair
 - Autoimmunity
 - Systemic inflammation
 - GALT dysfunction
 - Dermatitis, eczema
- Energy
 - Fatigue
 - Neurodegenerative disease
- Mitigation and Clearance
 - Liver inflammation, M A S L D
- Transport
 - Kidney disease
 - C V D
- Communication
 - Fertility challenges
 - M A S L D
 - Metabolic Disease: diabetes, insulin resistance, metabolic syndrome
- Structural integrity
 - Peptic ulcer, blunted villi
 - Osteopenia, osteoporosis
 - Sarcopenia
- Assimilation
 - DIG IN
 - Chronic Gastritis
 - Crohn's and Ulcerative Colitis
 - N SAID induced enteropathy

References

GUT INFLAMMATION AS MEDIATOR

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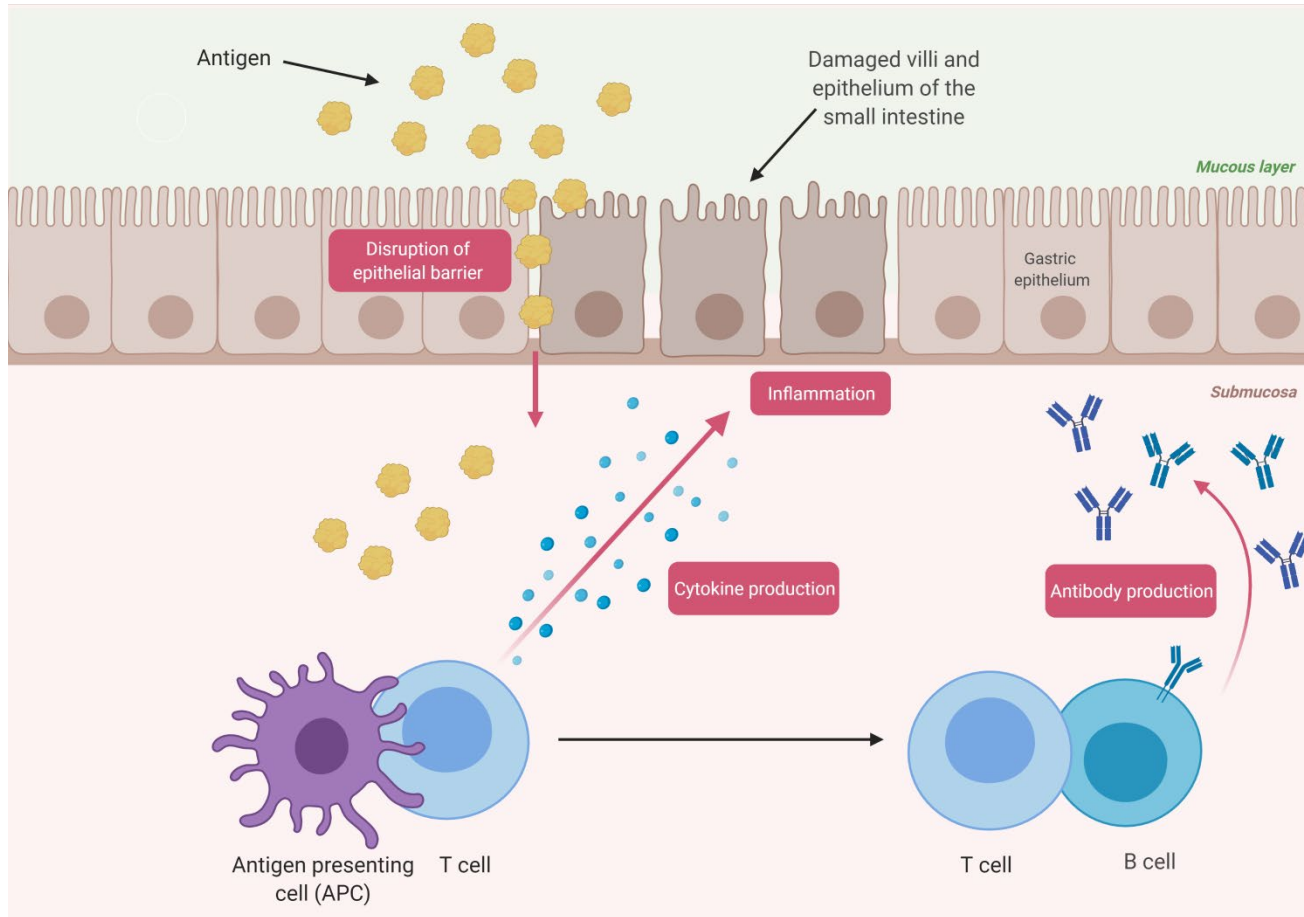
Gut Inflammation Mediates Multiple Gut-Organ Axis Disorders



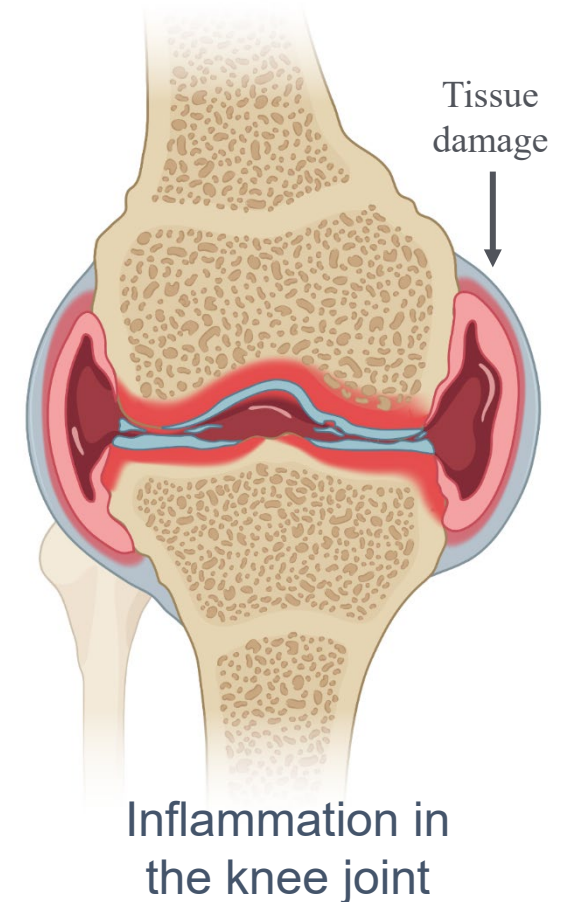
Long Description: Gut Inflammation

- Gut-skin axis
- Gut-bone axis
- Gut-liver axis
- Gut-heart axis
- Gut-pancreas axis
- Gut-muscle axis
- Gut-brain axis
- Gut-lung axis

Cross Reactivity Model of Autoimmunity: Role of Gut-Immune Modulation and Inflammation



The immune system confuses self with non-self. In this case, the immune system responds to cartilage in the knee joint.

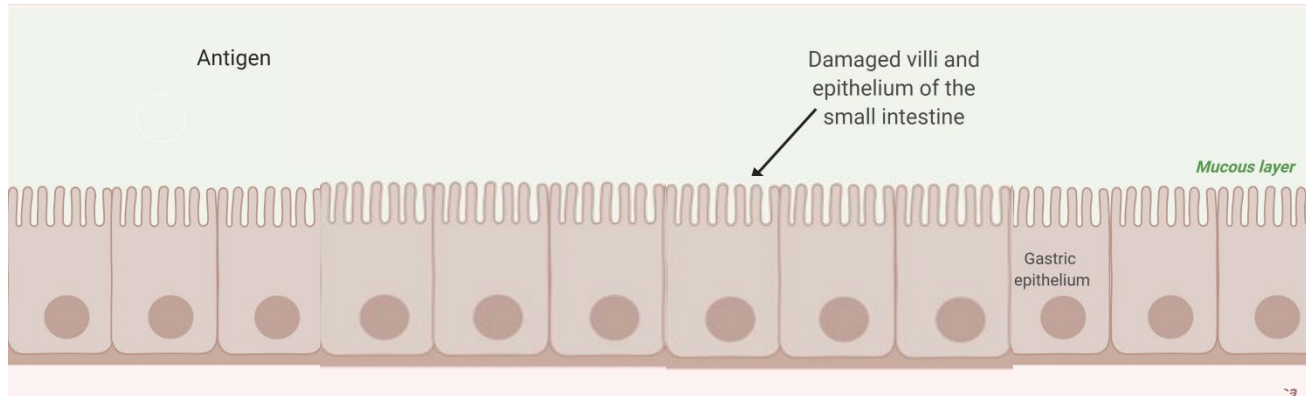


- Image: Adapted from "*H. Pylori* Pathogenesis" by BioRender.com (2020). Retrieved from <https://app.biorender.com/biorender-templates>
- See "References: Cross Reactivity Model of Autoimmunity"

Long Description: Cross Reactivity Model of Autoimmunity: Role of Gut-Immune Modulation and Inflammation

- In the presence of an impaired intestinal barrier and dysbiosis, a lack of immune tolerance towards dietary antigens leads to an altered immune response and an increased risk of inflammatory, autoimmune, and atopic conditions. An example of this is an immune response to cartilage in the knee joint, causing inflammation and tissue damage.

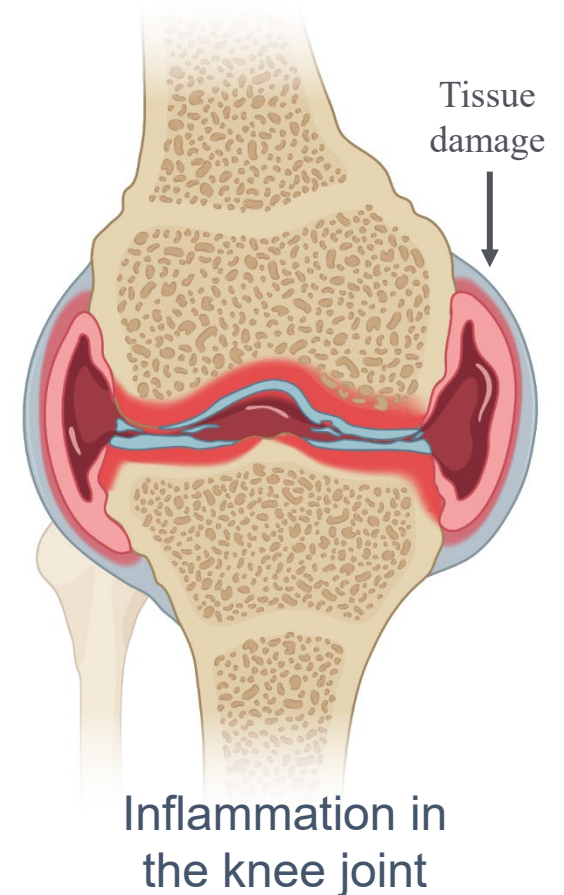
Cross Reactivity Model of Autoimmunity: Role of Gut-Immune Modulation and Inflammation



Food for NOURISHMENT
that is digested and ready for
ASSIMILATION



The immune system confuses self with non-self. In this case, the immune system responds to cartilage in the knee joint.



- Image: Adapted from “*H. Pylori* Pathogenesis” by BioRender.com (2020). Retrieved from <https://app.biorender.com/biorender-templates>
- See “References: Cross Reactivity Model of Autoimmunity”

Long Description: Cross Reactivity Model of Autoimmunity: Role of Gut-Immune Modulation and Inflammation

- With an intact gut barrier, food is digested, assimilated, and serves as nourishment.

References

CROSS REACTIVITY MODEL OF AUTOIMMUNITY

- Brusca SB, Abramson SB, Scher JU. Microbiome and mucosal inflammation as extra-articular triggers for rheumatoid arthritis and autoimmunity. *Curr Opin Rheumatol*. 2014;26(1):101-107. doi:10.1097/BOR.0000000000000008
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Healing Gut Inflammation

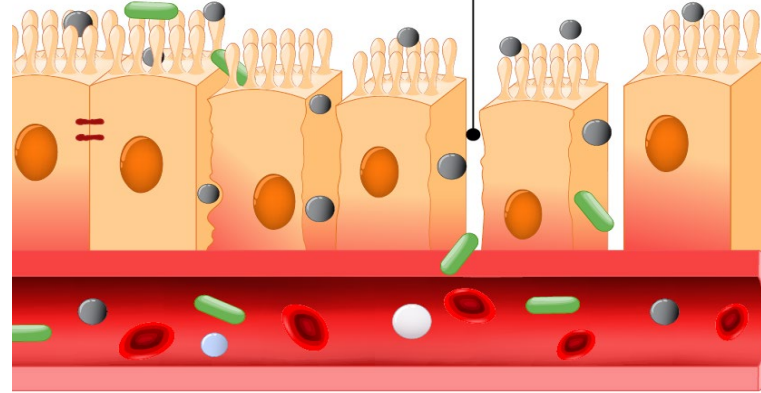


Image Courtesy of Swathi Rao, PA-C

ATMs and Mechanisms of Allergy and Autoimmunity

Food (gluten, casein, etc.)
Toxins Stress Dysbiotic Organisms

↑ Intestinal Permeability



Inflammatory Cytokines

Leakage of LPS and Other Inflammatory Factors

Allergic, Autoimmune, Systemic Inflammation



Long Description: A T Ms and Mechanisms of Allergy and Autoimmunity

- In the presence of an impaired intestinal barrier and dysbiosis, leakage of L P S and other inflammatory factors leads to an altered immune response and an increased risk of inflammatory, autoimmune, and atopic conditions, with inflammatory cytokines potentiating this process.



Hygiene Hypothesis → Epithelial Barrier Theory

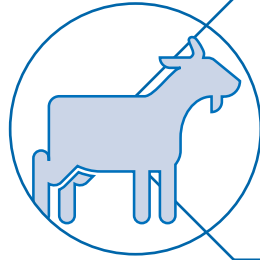
- The epithelial barrier theory (EBT) incorporates the hygiene and biodiversity hypotheses with epithelial barrier damage and GI dysbiosis (i.e., gut inflammation) to propose mechanisms for developing allergic, autoimmune, and inflammatory diseases.
- The EBT also suggests that exposure to harmful substances from the environment can disrupt intestinal epithelial barriers and contribute to dysbiosis, which highlights how a patient's environmental exposures may serve as mediators of gut inflammation.

Theories of Gut-Immune Modulation



Hygiene Hypothesis

- Proposed in 1989 by Strachan; relates excessive cleanliness and a lower incidence of childhood infection to increased incidence of allergic diseases and atopy.



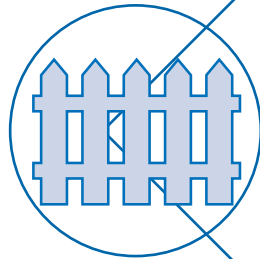
Old Friends Hypothesis (Immune Tolerance)

- Proposed in early 2000s by Rook; suggests that certain microbial strains that have evolved alongside humans are viewed as harmless (“old friends”) by the immune system.



Biodiversity Hypothesis

- Increased exposure to manifold environmental settings may increase microbial diversity in humans and provide a protective effect with regard to inflammatory ailments.



Epithelial Barrier Theory

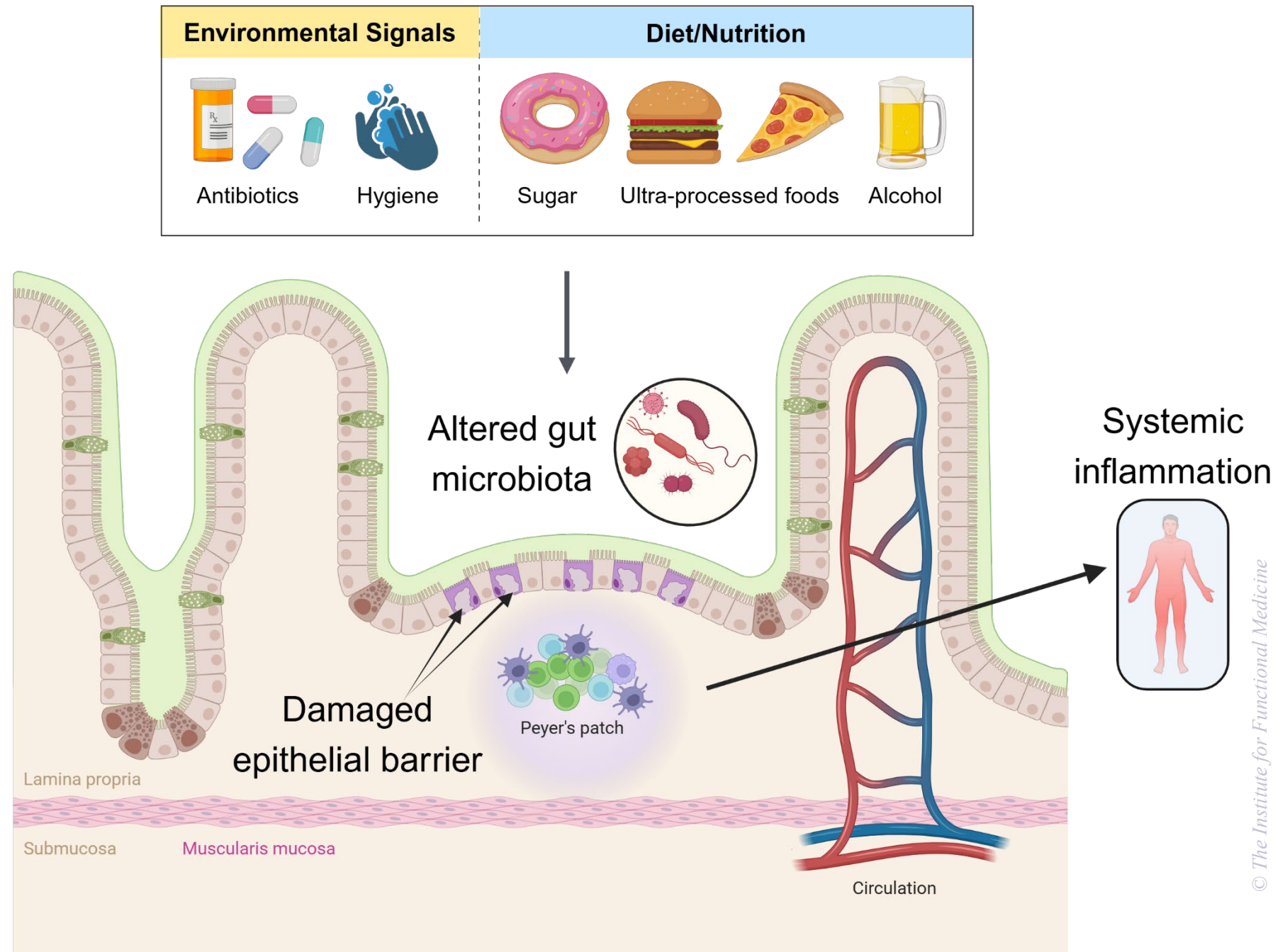
- Toxins and other harmful environmental insults may weaken the structural integrity of protective barriers (skin, lung, gut) and disrupt the balance of the microbiome.

Clinical Focus

- Summary, conclusion, and key clinical takeaways.



GALT & Gut Inflammation: The Fulcrum Between Inflammatory Mediators and Chronic Systemic Inflammatory Diseases

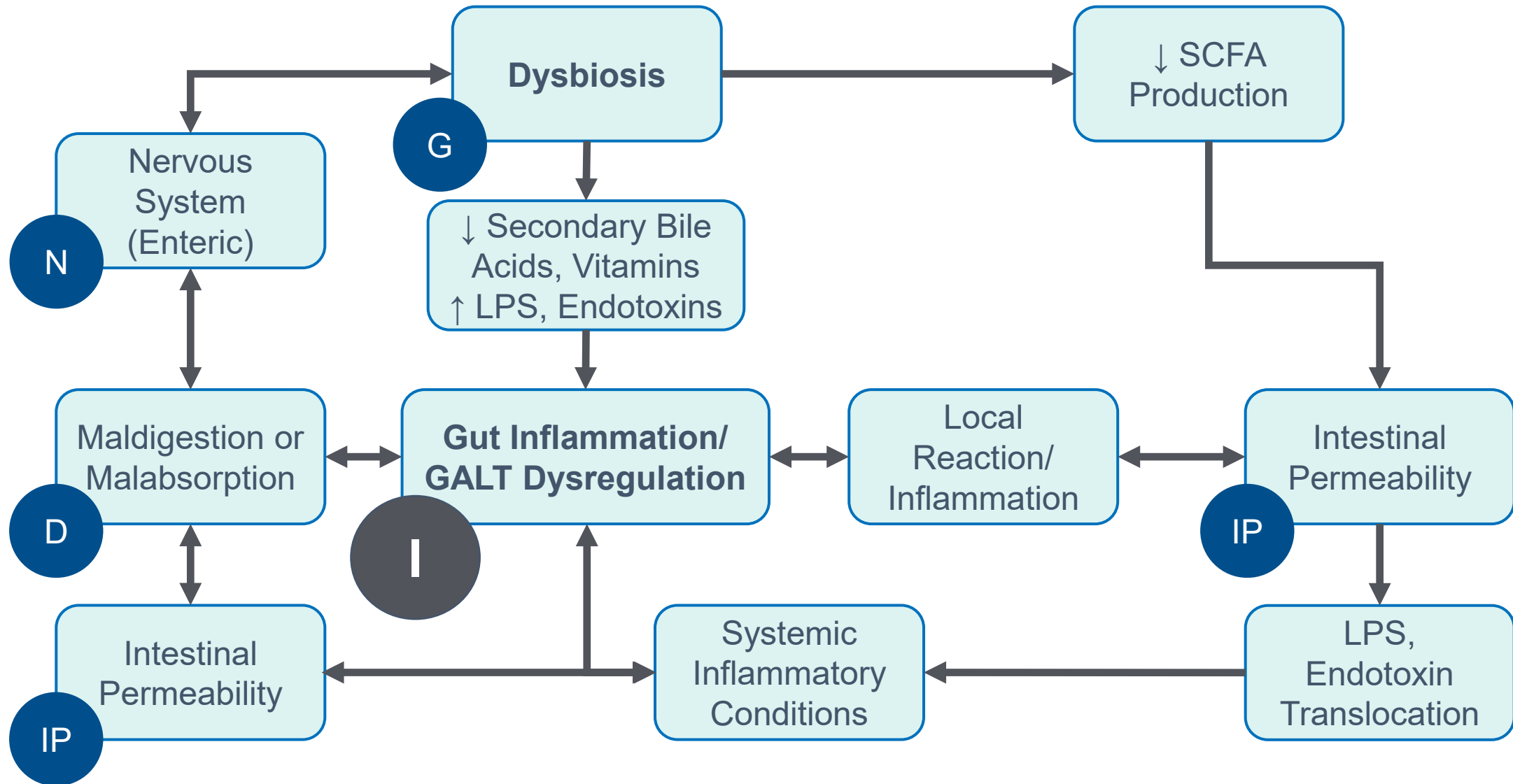


Long Description: G A L T and Gut Inflammation

- Environmental signals such as antibiotic use, hygiene practices, and dietary choices such as sugar, ultra-processed foods, and alcohol are mediators of systemic inflammation through altered gut microbiota and a damaged epithelial barrier.

Clinical Scenarios of Gut/GALT Inflammation

Correlation With DIGIN Dysfunctions

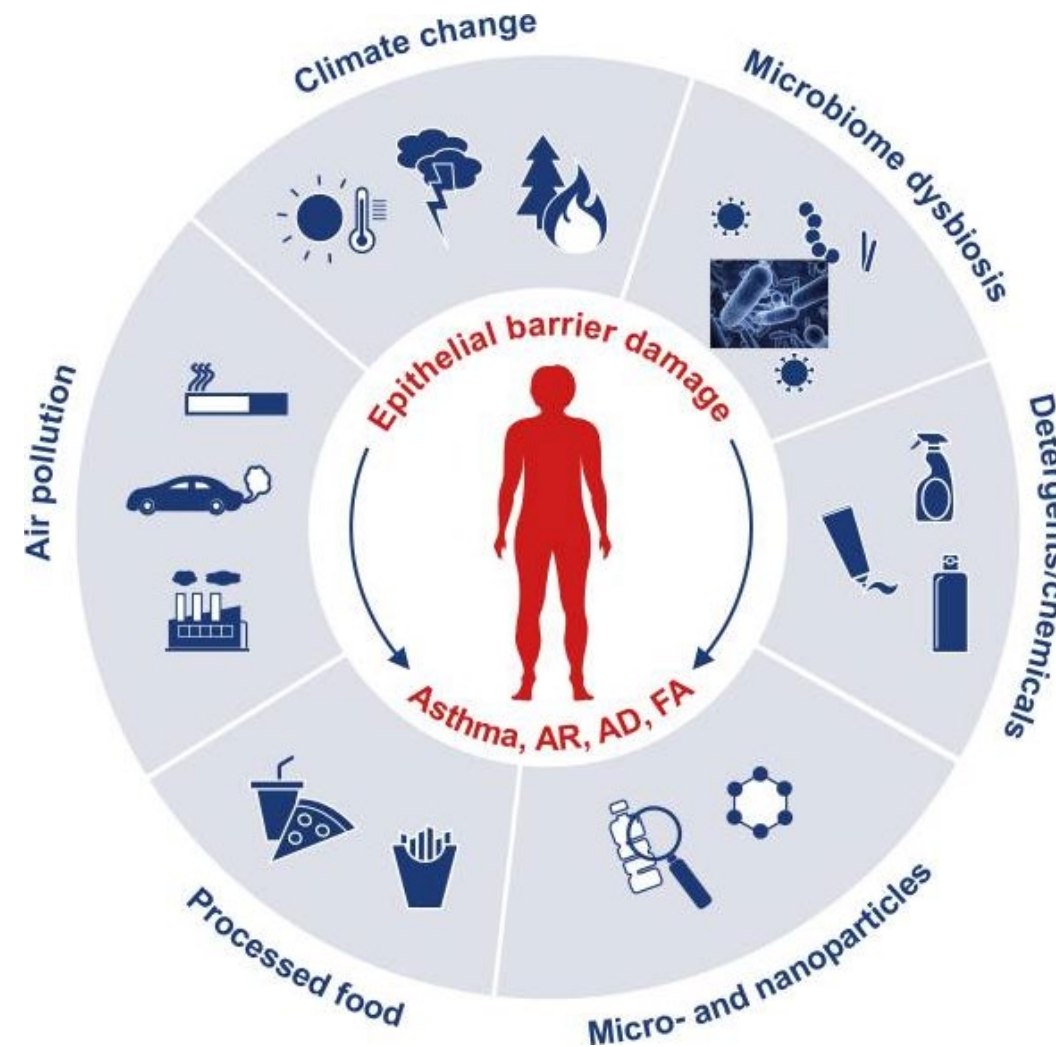


Long Description: Clinical Scenarios of Gut/G A L T Inflammation

- Recall the D I G I N dysfunctions and their mutual reinforcements that lead to systemic inflammatory conditions
 - D. Digestion, absorption
 - I. Intestinal permeability
 - G. Gut microbiota, dysbiosis
 - I. Inflammation, immune modulation
 - N. Nervous system, enteric division



21st Century “Emerging” Antecedents, Triggers, and Mediators Gut Inflammation

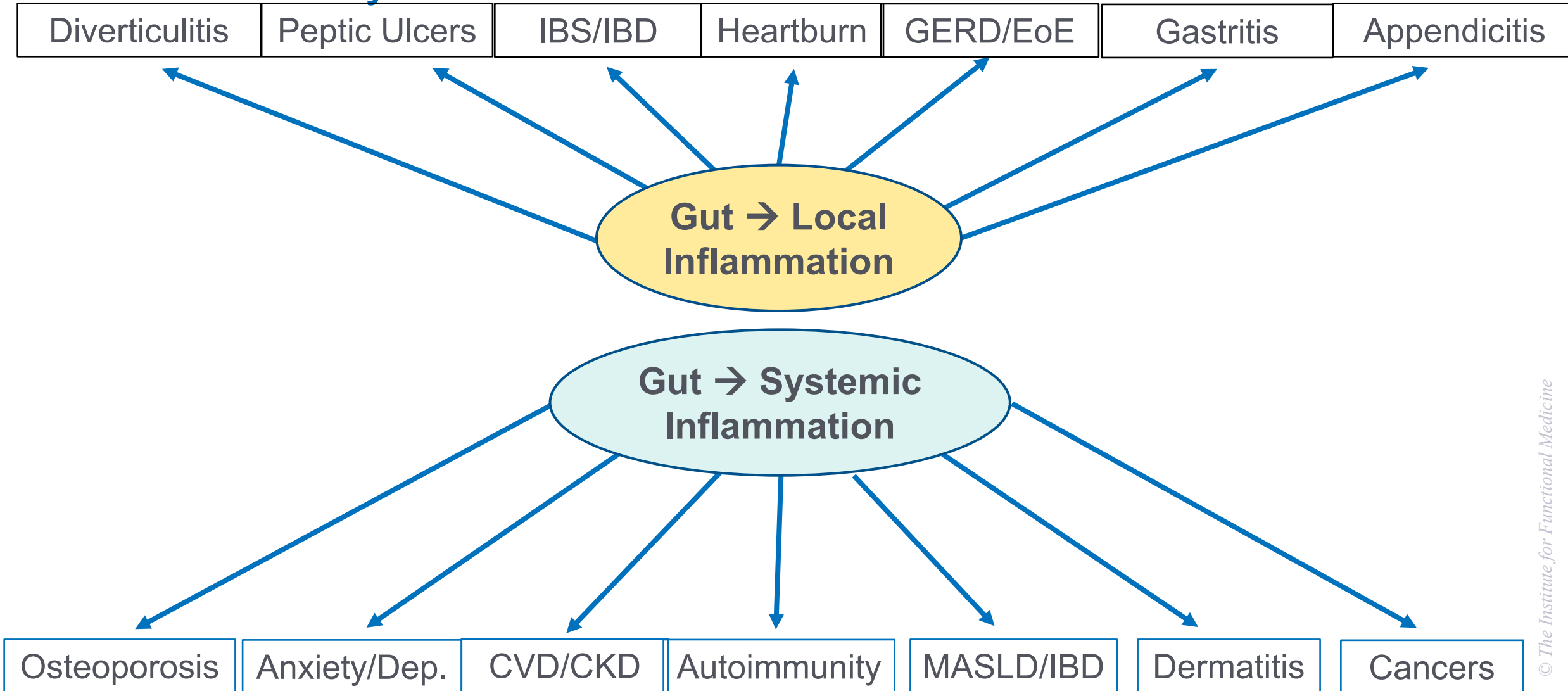


Long Description: 21st Century “Emerging” Antecedents, Triggers, and Mediators Gut Inflammation

- Climate change
- Microbiome dysbiosis
- Detergents and chemicals
- Micro and nanoparticles
- Processed food
- Air pollution

Gut Inflammation

Local and Systemic Manifestations



See: "References: Local and Systemic Manifestations of Gut Inflammation"

Long Description: Gut Inflammation Local and Systemic Manifestations

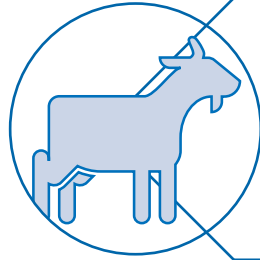
- Local gut inflammation manifestations
 - Diverticulitis
 - Peptic ulcers
 - I B S and I B D
 - Heartburn
 - GERD and E O E
 - Gastritis
 - Appendicitis
- Systemic gut inflammation manifestations
 - Osteoporosis
 - Anxiety, depression
 - C V D and C K D
 - Autoimmunity
 - M A S L D
 - I B D
 - Dermatitis
 - Cancers

Theories of Gut-Immune Modulation



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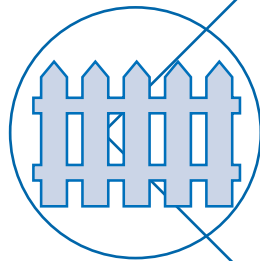
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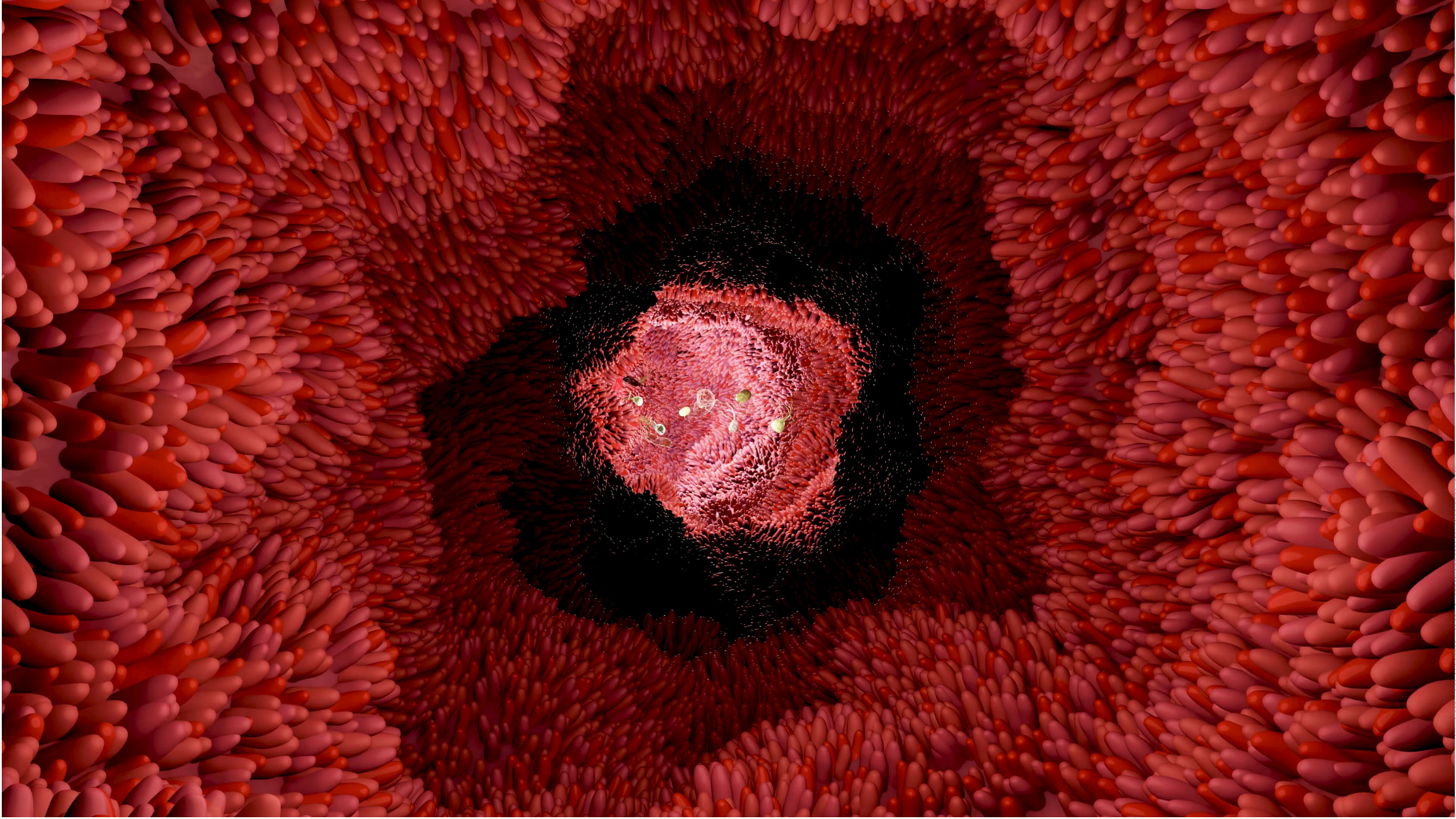
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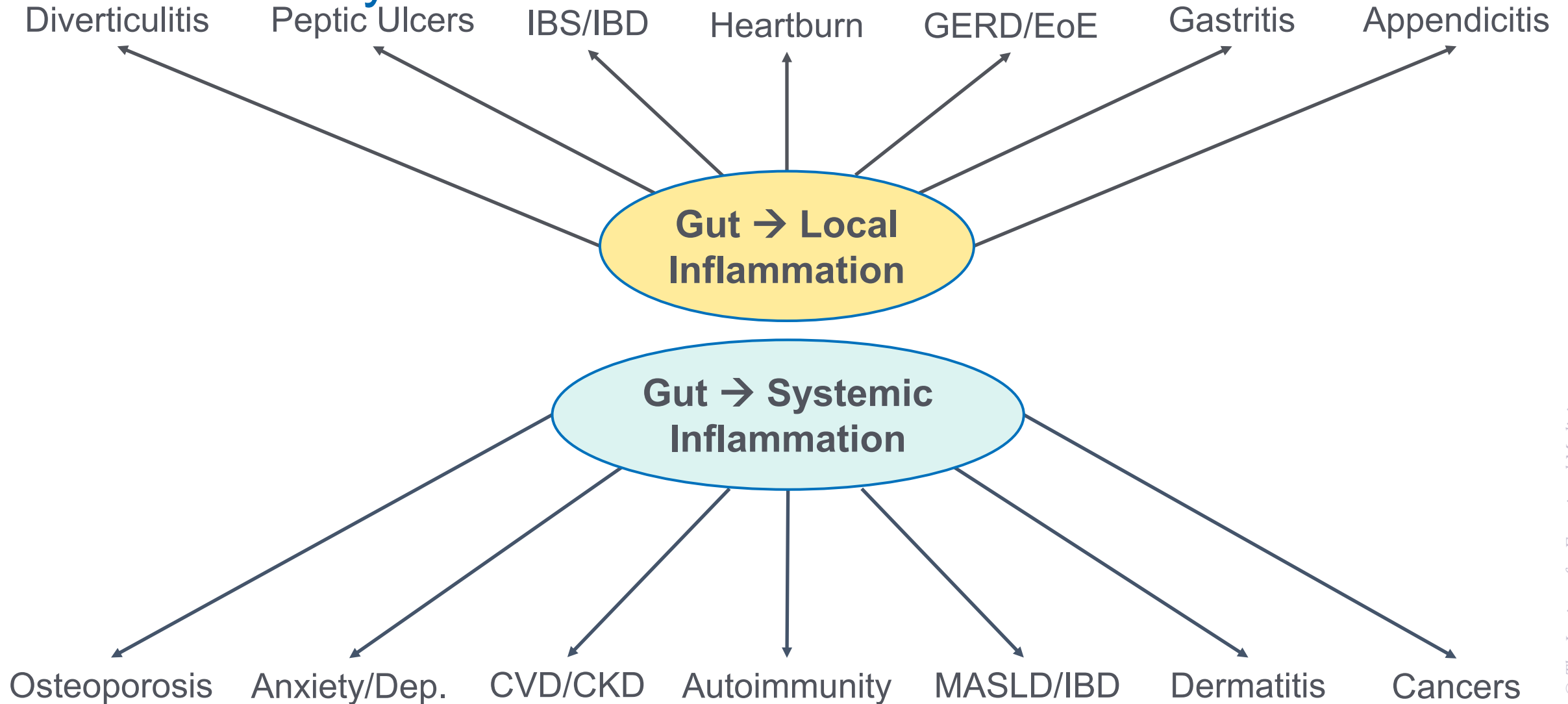


Long Description: The Intestinal Barrier

- The structure and function of the intestinal barrier are essential to absorption and immune function.

Gut Inflammation

Local and Systemic Manifestations



See: "References: Local and Systemic Manifestations of Gut Inflammation"

Gut Inflammation:

Manifestations – Mediators – Mitigators

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE